

SIMATS ENGINEERING — AI&DS Department

Edge Computing and Reliable Data Transmission

Course: Computer Networks (CSA0707)

Seminar Topic: Edge Computing and Reliable Data Transmission

Presented By: **M. VAISHNAV**

- Roll No: 192524251
- AI&DS Department

Date: July 25, 2026

Introduction to Edge Computing

Definition

Edge computing is a distributed computing model that brings computation and storage closer to where data is created.

Real-Time Processing

By processing data at the point of generation, edge computing avoids the delay of sending everything to the cloud.

Importance in Modern Networks

It supports fast decisions in IoT, 5G, and mission-critical systems where every millisecond matters.

Latency Advantage

Edge computing offers a major speed benefit, delivering much lower latency than traditional cloud processing.

Architecture of Edge Computing



End Devices:

IoT sensors, cameras, wearables, and mobile devices generate data at the source.

Edge Gateways

They filter, aggregate, and pre-process data before it moves further.

Local Processing:

This reduces latency and avoids sending every event to the cloud.

Cloud Data Centres:

They support large-scale storage, coordination, and deeper analysis.

Reliable Data Transmission

Definition

Reliable transmission means data sent from the source reaches the destination completely, correctly, and in order.

It is the foundational contract of dependable networking.

Error Detection

Parity bits, Cyclic Redundancy Check (CRC), and checksums help identify corrupted packets in transit.

They prevent invalid data from being accepted by the receiver.

Data Integrity

Hash functions such as MD5 and SHA verify that received data matches the original.

This helps prevent silent corruption that could affect application logic.

Role of Edge Computing in Reliable Data Transmission

Reduced Network Congestion

By filtering data at the source, edge computing lowers traffic volume and reduces packet collisions. This helps prevent data loss before information reaches the backbone network.

Faster Response Time

Local error correction at the edge avoids long-haul round trips to the cloud. As a result, recovery latency is reduced and responses happen much faster.

Improved Quality of Service

Edge nodes can prioritise critical traffic such as medical alerts and vehicle control signals. This keeps important data moving reliably even when networks are busy.

Conclusion

Edge Computing Benefits

Reduces latency by processing data closer to the source.
Also lowers bandwidth usage by minimizing unnecessary cloud traffic..

Future Relevance

They are essential for IoT ecosystems and smart devices.
They also support the growth of 5G network deployments.

Key Takeaway

The seminar highlights the need for fast, reliable networking.
Edge computing and transmission methods are key enablers.

Thank You

M. VAISHNAV — Roll No: 192524251
AI&DS Department, SIMATS ENGINEERING
Computer Networks (CSA0707)